

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OPERATING SYSTEMS COURSE CODE: CSA0401

COURSE OUTCOME COVERED

C02: Analyze process management and deadlock handling techniques to improve CPU utilization and system performance(BL3-BL4)

Assessment Tool Weightage: 30%

QUESTIONS: 10

Total Marks:50

Annexure A

Scenario based

Code of Conduct:

I ___G. Yashmita(192511032)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Yashmita

Question 1 – CPU Scheduling in a Real-Time Ticket Booking System

1. Suitable Scheduling Algorithm

The most suitable scheduling algorithm is **Priority Scheduling** because payment transactions are the most important and should be processed first.

Priority Order:

- **P1 – Payment Gateway (High)**
- **P2 – Seat Availability (Medium)**
- **P3 – Ticket Confirmation (Low)**

2. Justification

- High-priority payment requests are executed immediately.
- Seat availability is checked after successful payment.
- Ticket confirmation is processed last.
- This reduces customer waiting time and improves CPU utilization during heavy traffic.

3. Sample Execution Order

Process	Priority	Burst Time
P1	High	5 ms
P2	Medium	8 ms
P3	Low	10 ms

Execution Order:

P1 → P2 → P3

Gantt Chart

0	5	13	23
P1	P2	P3	

4. Drawback

Priority Scheduling may cause **starvation**, where low-priority processes wait for a long time if high-priority processes keep arriving.

Solution: Use **Aging**, which gradually increases the priority of waiting processes.

Conclusion

Priority Scheduling is best because it executes important tasks first, reduces waiting time, and improves CPU utilization. Starvation can be prevented using Aging.

Question 2 – Deadlock in a Banking Transaction Server

1. Text-Based Resource Allocation Graph

$A \rightarrow T1 \rightarrow B \rightarrow T2 \rightarrow C \rightarrow T3 \rightarrow A$

This forms a circular chain.

2. Deadlock Identification

Yes, a deadlock has occurred.

- T1 holds A and waits for B.
- T2 holds B and waits for C.
- T3 holds C and waits for A.

Since each transaction waits for another, none can proceed.

3. Technique to Fix the Issue

Use **Deadlock Prevention** by requesting resources in a fixed order:

$A \rightarrow B \rightarrow C$

This removes the circular wait condition and prevents deadlock.

4. Locking Order

1. Lock A
2. Lock B
3. Lock C
4. Perform transaction
5. Release C
6. Release B
7. Release A

This avoids deadlock and improves system performance.

Conclusion

The banking server enters a deadlock because of circular waiting. Using a fixed locking order prevents future deadlocks.

Question 3 – Mobile OS App Switching Lag

1. Why CPU Remains Idle

Many background apps are waiting for I/O operations such as network or disk access. Since they are not ready to execute, the CPU becomes idle even though many apps are active.

2. Is the 20 ms Quantum Suitable?

No. A **20 ms time quantum** is too large for interactive mobile applications.

It causes:

- Slow app switching
- Increased waiting time
- Poor user experience

3. Better Scheduling Method

Use **Priority Scheduling** or reduce the Round Robin time quantum to **10 ms**.

This allows foreground apps like WhatsApp to receive CPU time faster.

4. Benefits

- Faster app switching
- Reduced response time
- Better CPU utilization
- Improved system throughput

Conclusion

Using Priority Scheduling or a smaller time quantum improves responsiveness, reduces lag, and increases overall system performance.

Question 4 – Deadlock Risk in an Operating System Update

1. Can Deadlock Occur?

Yes.

- M1 holds R1 and waits for R2.
- M2 holds R2 and waits for R3.
- M3 holds R3 and waits for R1.

This creates a deadlock.

2. Circular Wait

M1 → R2 → M2 → R3 → M3 → R1 → M1

Each module waits for a resource held by another module.

3. Banker's Algorithm

Allocate resources only if the system remains in a **safe state**.

Safe sequence:

M1 → M2 → M3 → M4

This prevents deadlock.

4. Module Priority

Module	Priority
M1 – Copy Files	High
M2 – Install Drivers	High
M3 – Update Registry	Medium
M4 – Validate System	Low

This improves CPU utilization and reduces waiting.

Conclusion

The update process can lead to a deadlock due to circular waiting. Banker's Algorithm and proper module priority prevent deadlocks and improve performance.

Question 5 – Cloud Server CPU Utilization Analysis

1. CPU Utilization Calculation

Given:

- $p = 0.8$
- $n = 7$

Formula:

$$U = 1 - p^n$$

Calculation:

$$U = 1 - (0.8)^7$$

$$U = 1 - 0.2097 = 0.7903$$

CPU Utilization = 79.03%

2. Interpretation

A CPU utilization of **79.03%** indicates that the server is **optimally loaded**, although some time is lost due to I/O waiting.

3. Effect of Number of VMs

- Increasing VMs improves throughput until the CPU becomes overloaded.
- Decreasing VMs may increase CPU idle time and reduce throughput.

4. OS-Level Strategies

1. Use efficient CPU scheduling (Priority Scheduling or MLFQ).
2. Reduce I/O waiting using caching, buffering, or asynchronous I/O.

Conclusion

The server has **79.03% CPU utilization**, indicating efficient usage. Proper scheduling and reducing I/O waiting can further improve performance and throughput.

RUBRICS FOR CALCULATION

Scenario based assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand